

Class Test-2

There are four sections. Each section is compulsory.

Total time is 45 minutes.

Section-1: (Attempt any four questions)

Q1: A data centre sends packets through a router. Each packet has a 10% probability of being corrupted, independently of others. In one batch, 8 packets are transmitted. Let X = Number of corrupted packets.

Tasks:

- What is the probability that exactly 2 packets are corrupted?
- What is the probability that at most 1 packet is corrupted?
- Find the mean and variance of corrupted packets.

Q2: A software module receives, on average, 3 bugs per day.

Let X = Number of bugs per day

Tasks:

- Probability of exactly 5 bugs in a day?
- Probability of no bugs?
- Mean and variance?

Q3: A cloud server response time (in seconds) follows the PDF:

$$f(x) = \frac{1}{5}e^{-x/5}, x > 0$$

This is an Exponential distribution with:

$$\lambda = \frac{1}{5}$$

Tasks:

- What is the probability response time exceeding 8 seconds?
- What is the probability response time is less than 3 seconds?
- Find mean and variance.

Q4: CPU usage percentage is modelled as:

$$f(x) = kx^2, 0 < x < 2$$

Tasks:

- Find k .
- Probability CPU usage is between 1 and 2.
- Find mean.

Q5: A machine learning classifier misclassifies an image with probability 0.15. A batch of 12 images is tested independently.

Let X = Number of misclassified images

Tasks:

- What is the probability that exactly 3 images are misclassified?
- What is the probability that at least 2 are misclassified?
- Find mean and variance.

Q6: A server experiences 4 cyber-attacks per hour on average.

Let X = Number of attacks per hour

Tasks:

- Probability of exactly 6 attacks?
- Probability of fewer than 2 attacks?
- Mean and variance?

Q7: The lifetime (in months) of a cloud instance follows an exponential distribution with a mean = 10 months.

Thus, $\lambda = \frac{1}{10}$

Tasks:

- Probability server lasts more than 15 months.
- Probability it fails within 5 months.
- Find variance.

Section-2 : (Attempt any two questions)

Q1: A company tests 10 network devices per day. Each device may be defective or not. Over 200 days, the following frequency of defective devices was recorded:

No. of Defectives (x)	Frequency (f)
0	38
1	62
2	54
3	30
4	16

Tasks:

- Find the sample mean number of defectives using the expectation formula.
- Estimate the probability of defect (p).

- iii. Find the theoretical mean and variance assuming Binomial.

Q2: A server records the number of login failures per hour. Data for 300 hours:

Failures (x)	Frequency (f)
0	40
1	75
2	90
3	55
4	30
5	10

Tasks:

- Find the mean using the expectation formula.
- Estimate λ .
- Find the probability of exactly 3 failures using Poisson.

Q3: A technical support centre recorded waiting time (in minutes) for 100 customers:

Time Interval	Frequency
0-2	30
2-4	25
4-6	20
6-8	15
8-10	10

Tasks:

- Estimate mean waiting time.
- Estimate λ .
- Find the probability that the waiting time exceeds 6 minutes.

Section-3: (Attempt any two)

Q1: A network transmits 5 packets; each packet fails with probability 0.2 independently.

Let X = Number of failed packets

So, $X \sim \text{Binomial}(n = 5, p = 0.2)$

Tasks:

- i. Using MGF.
- ii. Using Probability Generating Function (PGF).

Q2: A cloud server experiences, on average, 3 bugs per day.

$$X \sim \text{Poisson}(\lambda = 3)$$

Tasks:

- i. MGF.
- ii. PGF.

Q3: A server response time follows an exponential distribution with a mean of 4 seconds.

$$\lambda = \frac{1}{4}$$

Tasks:

- i. MGF.
- ii. Characteristic Function (CF).

Section-4: (Attempt all)

Q1: A cybersecurity monitoring system records the number of intrusions attempts per hour over 200 hours. The observed frequency data is:

No. of Intrusions (x)	Frequency (f)
0	30
1	50
2	60
3	40
4	20

You are required to:

- i. Find mean and variance from frequency data.
- ii. Identify the suitable probability distribution.
- iii. Derive the MGF.
- iv. Find the mean using MGF.
- v. Write the PGF.
- vi. Write the Characteristic Function.

Q2: A cloud data centre records server response time (seconds) for 1000 requests:

Time Interval	Frequency
0-2	400
2-4	250
4-6	180
6-8	100
8-10	70

Tasks:

- i. Estimate mean and variance.
- ii. Identify distribution.
- iii. Write MGF.
- iv. Write CF.
- v. Verify mean using MGF.